

Ideation Phase

Empathize & Discover

Date	13 July 2026
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes. It helps teams better understand their users.

Creating an effective solution requires understanding the true problem and the person experiencing it. Building this map helped the team consider the housing market from the user's own perspective, along with their goals and challenges.

Our Empathy Map Canvas: Home Buyer

Primary user persona: a prospective home buyer comparing multiple property listings, trying to judge whether the asking price of a house is fair given its size, condition, age and location.

THINKS • Wonders whether the asking price genuinely reflects the size, condition and grade of the house, or if it is overpriced for the area. • Weighs whether paying a premium for a renovated home or a waterfront view is worth the extra cost. • Questions how much the age of the house will affect future maintenance and repair expenses. • Considers whether a smaller home in a better zipcode is a smarter buy than a larger home further out.	SEES • Sees very different sale prices for houses with a similar number of bedrooms and bathrooms. • Notices that some listings highlight recent renovations while others show no renovation history at all. • Observes that homes in certain zipcode groups consistently list higher than others. • Sees inconsistent grading and condition ratings (Excellent, Good, Fair, Okay) across otherwise similar homes.
SAYS • “I want to know if this price matches similar homes nearby before I make an offer.” • “A renovated home should genuinely be worth more — but by how much?” • “Living area and lot size matter more to me than just the bedroom count.” • “I’d pay more for a waterfront view, but only if the rest of the price makes sense.”	DOES • Compares sale price, living area, lot area and grade across several listings side by side. • Checks each house's age, renovation history and years since renovation before shortlisting. • Filters listings by zipcode group and location to stay within a preferred neighbourhood. • Weighs waterfront view and overall condition against the final asking price.
PAINS • Frustrated when price does not appear to align with the size, condition or features of the home. • Finds it tedious to manually compare dozens of listings across many features at once.	GAINS • Values a transparent, side-by-side comparison of price against house features. • Wants confidence that the price paid genuinely reflects the home's true condition and location value.

• Unsure whether a renovation genuinely adds long-term value or is just a marketing point. • Concerned about hidden long-term costs tied to an older, un-renovated house.	• Appreciates seeing how renovation, age and waterfront view actually move the sale price. • Enjoys being able to filter and compare homes quickly by budget and preferred features.
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User Needs

- A clear view of how price relates to core features — bedrooms, bathrooms, living area and lot area — rather than a single asking figure.
- A way to see whether renovation status and years since renovation genuinely affect resale value.
- Visibility into how location (zipcode group) influences typical sale prices in an area.
- Confidence that a home's grade, condition and age are fairly reflected in its price.

Opportunities

- Use historical sales data to show how price varies with living area, bedrooms, bathrooms and overall grade.
- Segment and visualize sale prices by zipcode group to reveal location-driven price patterns.
- Correlate renovation status and years since renovation with sale price to quantify a “renovation premium.”
- Highlight the price impact of waterfront view and house age to help buyers judge long-term value.